

# A Large-Scale Dataset of Distributed Renewable Energy (DDRE-33)

Statistical Methodology & Validation

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|-------------|-------------|-------------|
| Saahitthi   | Drishti     | Shreya      |
| 2022IMT-011 | 2022IMT-040 | 2022IMT-053 |
| Priya       | Pravallika  | Muskaan     |
| 2022IMT-061 | 2022IMT-062 | 2022IMT-073 |

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## Problem Statement

How can we transform a raw, noisy, and high-dimensional distributed renewable energy dataset into a statistically consistent, reliable, and interpretable dataset using systematic research methodologies such as hypothesis testing, anomaly detection, and dimensionality reduction?

## Literature Review

- Modern power systems increasingly rely on **distributed renewable energy (DRE)**, introducing high variability and uncertainty. [1] [2] 36
- Advanced forecasting and optimization methods using **deep learning and multi-energy systems** have been widely explored. [3] [4]
- Standard test systems like the **IEEE-33 bus network** provide a benchmark for distribution studies but lack realistic renewable datasets. [5]
- Existing datasets either focus on **centralized generation** or lack **spatiotemporal correlations and labeling**, limiting their usability. [6]

## Research Gap

From the literature review, it is evident that there is a lack of datasets specifically designed for distributed renewable energy systems. Existing datasets do not adequately model spatial and temporal correlations and often suffer from missing values and lack of labeling. Additionally, integrated wind and solar scenarios are not sufficiently represented. This highlights a clear research gap, motivating the need for a statistically robust and well-structured dataset.

## Base Paper

- Proposes DDRE-33 dataset for distributed renewable energy systems
- Uses statistical preprocessing and synthetic data generation
- Validates data using correlation and distribution-based methods [7]

## DDRE-33 Dataset

- A large-scale dataset for distributed renewable energy scenarios on the IEEE-33 bus system [7]
- Captures spatial and temporal variations in wind and solar generation
- Publicly available dataset released by the authors via repository [8]

## Outlier Detection Methods Used

- Interquartile Range (IQR) and Isolation Forest

## Interquartile Range (IQR) Formula

$$IQR = Q_3 - Q_1 \quad (1)$$

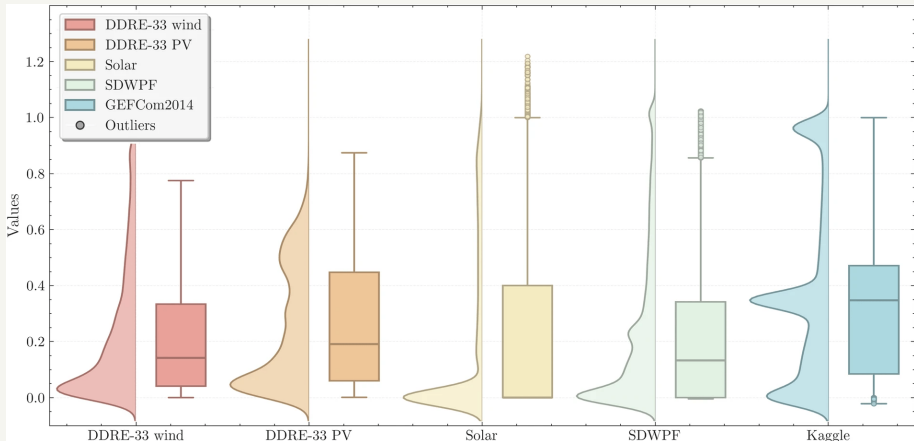
$$\text{Lower} = Q_1 - 1.5 \times IQR \quad (2)$$

$$\text{Upper} = Q_3 + 1.5 \times IQR \quad (3)$$

## Impact

- **Noise Reduction:** Removes extreme anomalies.
- **Model Performance:** Improves overall prediction accuracy.

# Data preprocessing Results



- Successfully identified data points outside statistical bounds.
- Cleaned dataset shows tighter variance and more realistic distribution.

## Missing Data Handling Methods Used

- Linear interpolation and Time-based averaging

## Mathematical Formulation

$$x_t = x_{t_1} + \frac{(x_{t_2} - x_{t_1})}{(t_2 - t_1)} \cdot (t - t_1) \quad (4)$$

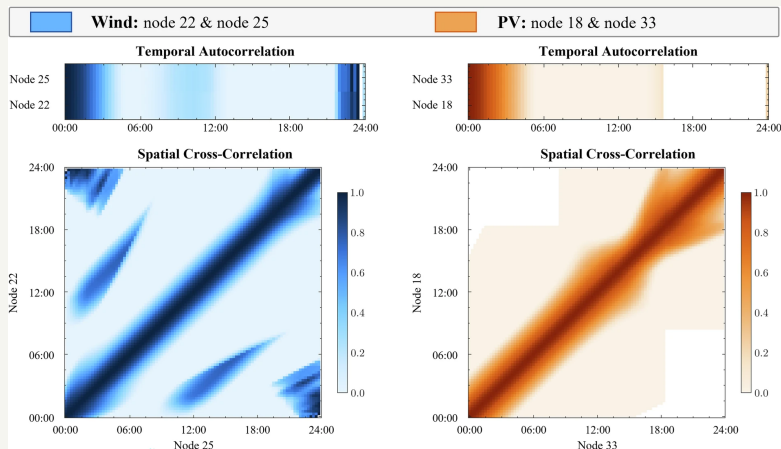
$$x_t = \frac{1}{n} \sum_{i=1}^n x_i \quad (5)$$

## Impact

- **Continuity:** Maintains time-series flow.
- **Integrity:** Prevents distortion in temporal analysis.



# Correlation Analysis Results



- Improves realism of generated renewable energy profiles.
- Ensures consistency across different bus locations.

# Hypothesis Testing: PV Node 18 vs PV Node 33



**Objective:** Compare solar (PV) power output across two nodes.

## Dataset Details:

- PV Node 18 vs PV Node 33
- Samples: 17,520,000 (each)

## Statistical Methods:

- Two-sample t-test (mean comparison)
- F-test (variance comparison)

## Hypotheses:

### T-test:

- $H_0: \mu_{18} = \mu_{33}$
- $H_1: \mu_{18} \neq \mu_{33}$

### F-test:

- $H_0: \sigma_{18}^2 = \sigma_{33}^2$
- $H_1: \sigma_{18}^2 \neq \sigma_{33}^2$

**Significance Level:**  $\alpha = 0.05$

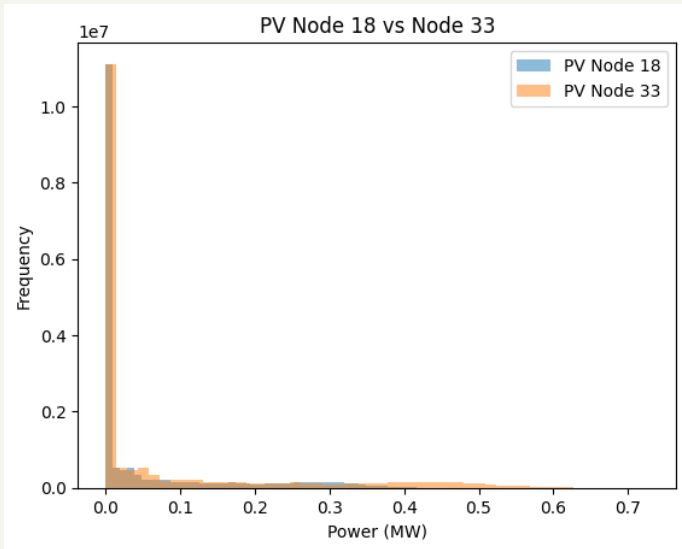
## Descriptive Statistics:

- Mean PV18: 0.05478 MW      Mean PV33: 0.08215 MW
- Var PV18: 0.00991      Var PV33: 0.02227

## Statistical Tests:

- T-test:  $T = -638.60$ ,  $P\text{-value} < 0.001 \Rightarrow \text{Reject } H_0$
- F-test:  $F = 0.4447$ ,  $P\text{-value} < 0.001 \Rightarrow \text{Reject } H_0$

# Results: PV Node 18 vs PV Node 33



## Inference:

- PV33 produces higher mean power than PV18
- Variability is also significantly higher at Node 33
- Results confirm statistical significance in both mean and variance

## Conclusion:

- Mean power outputs differ significantly (Reject  $H_0$  in t-test)
- Variability differs significantly (Reject  $H_0$  in F-test)
- Node 33 produces higher solar power than Node 18
- Node 33 exhibits greater variability (more fluctuation)
- Node 18 shows lower output but relatively more stable behavior

# Wind Node 22 vs Node 25: Hypothesis & Method



**Objective:** Compare wind power output across two nodes.

**Dataset:**

- Node 22 vs Node 25 (Wind)
- Samples: 17,520,000 each

**Statistical Tests:**

- Two-sample t-test (mean comparison)
- F-test (variance comparison)

**Hypotheses:**

**T-test:**

- $H_0: \mu_{22} = \mu_{25}$
- $H_1: \mu_{22} \neq \mu_{25}$

**F-test:**

- $H_0: \sigma_{22}^2 = \sigma_{25}^2$
- $H_1: \sigma_{22}^2 \neq \sigma_{25}^2$

**Significance Level:**  $\alpha = 0.05$

## Descriptive Statistics:

- Mean W22: 0.03989 MW
- Mean W25: 0.02981 MW
- Var W22: 0.00182
- Var W25: 0.00096

## T-Test:

- T-stat: 799.91
- P-value: 0.0

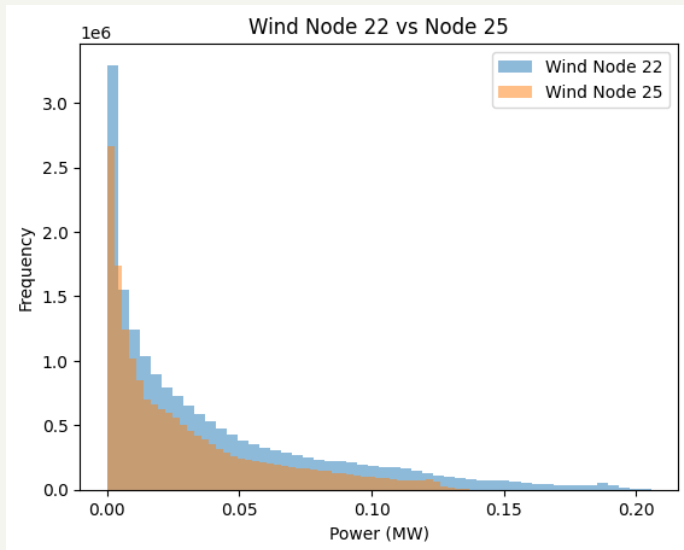
## F-Test:

- F-stat: 1.9048
- P-value:  $\approx 0$

## Decisions:

- Reject  $H_0$  (T-test)
- Reject  $H_0$  (F-test)

# Wind Distribution: Node 22 vs Node 25





## Inference:

- Node 22 distribution extends further right → higher power output
- Node 25 concentrated at lower values → lower generation
- Both distributions are right-skewed
- Node 22 shows wider spread → higher variability

## Conclusion:

- Node 22 is more productive but slightly more fluctuating
- Node 25 is less productive but more stable

## Principal Component Analysis (PCA)

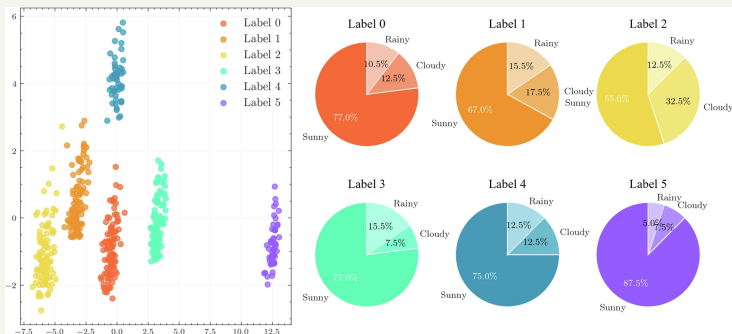
$$Z = XW \quad (6)$$

## K-means Clustering

$$\min \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2 \quad (7)$$

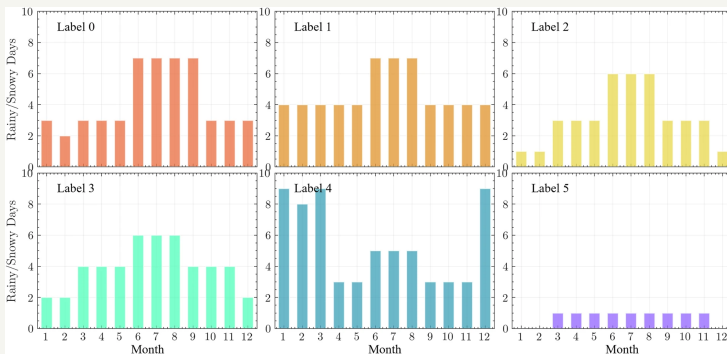
*Groups data into clusters based on similarity and projects variance.*

# Pattern Analysis Results



- Reveals hidden structures in distributed energy generation.
- Supports better decision-making and modeling.

# Climate Distribution



- Developed a statistically reliable dataset.
- Ensured data quality and consistency through pre-processing.
- Validated using multiple robust statistical methods.

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- [3] Y. G. et al., “Bilstm multitask learning-based combined load forecasting,” *IEEE Transactions on Smart Grid*, vol. 13, no. 5, pp. 3481–3492, 2022.
- [4] W. Z. et al., “Spatial–temporal graph neural network for multi-energy load forecasting,” *IEEE Transactions on Smart Grid*, vol. 15, pp. 2652–2666, 2023.
- [5] S. H. D. et al., “An enhanced ieee 33-bus benchmark test system for distribution system studies,” *IEEE Transactions on Power Systems*, vol. 36, pp. 2565–2572, 2020.
- [6] A. B. et al., “Multi-voltage level active distribution network with renewable integration,” *IEEE Transactions on Power Systems*, vol. 37, pp. 4874–4884, 2022.
- [7] Y. Chen, H. Xie, W. Huang, and P. Li, “A large-scale dataset of distributed renewable energy scenarios on the ieee-33 bus network,” *Scientific Data*, vol. 13, no. 146, 2026.



- [8] Y. Chen and H. Xie, "Ddre-33-chime: Code for ddre-33 dataset," <https://github.com/YuxuanCEE/DDRE-33-CHIME>, 2025, accessed: 2026-04-02.

# Thank You